Verify Important Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected claim-and-quantity audit.

Identify facts, repair units, inspect source origins and complete the new mass audit before reviewing real-world losses.

Keep one habit: Keep each claim, source, calculation and unresolved question together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Prompts Are Design Instructions

Privacy by Choosing Less

Good Data, Bad Data

W1 · Separate stated facts from derived quantities

> Use the supplied R reference and identify the evidence for each claim.

Your evidence record

- Card R reference for W1–W3: four rolls, exactly 2 m each; twelve pieces, exactly 50 cm each; no cutting loss in the paper model. Use 1 m=100 cm. R supplies no enjoyment evidence. Old Q concerns another activity with three rolls. Keep R separate from the 150 cm bookmark rolls and independent mass card M.

- Using R, identify a directly stated fact and a quantity that must be calculated. Give the **source** or expression for each. Explain why a **citation** to R does not verify the **claim** that every participant will enjoy the activity.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Repair the unit and check by a second route

> Convert consistently before subtraction and retain the model assumptions.

Your evidence record

- A report calculates 12 × 50 = 600 and labels the result metres. Use R to correct the **unit**, convert the result to metres and calculate the remaining length.

  Give a second arithmetic route or a check using starting, used and remaining lengths.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Inspect copied claims and missing citations

> Count independent evidence, not repeated posters.

Your evidence record

- Three posters say 600 metres were used, but all copied one draft. A fourth poster cites card Z, which is unavailable. Explain why these posters do not settle the quantity.

  Name the available **evidence** and check you would use instead.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Audit a new model with mass units

> Keep one first attempt before hints or feedback.

Read and explain independently

Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss.

A draft says “M supplies 1250 g; the models use 900 g; 350 kg remains; every model will win a prize.” Check all four **claim**s, show the remaining mass in grams, and explain the status of the prize **claim**. Use 1 kg = 1000 g.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Audit a new model with mass units

MAIN LESSON · SECTION 16

W5 · Review the mass model’s real-world limit

> After W4, distinguish a checkable no-loss calculation from an unsupported waste claim.

Your evidence record

- Another draft **claim**s that the amount left in W4 accounts for all real-world waste, citing a missing instruction card. Explain which result can still be checked from M and which **claim** remains unresolved.

  What information would be needed before revising the no-loss model?

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week’s change to your existing project record.

Your evidence record

Require human confirmation for important sorter **output**s and attach the **evidence** record used to approve each rule or label.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Repair a **unit** error for Verifying **claim**s and quantities. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Verifying **claim**s and quantities. Proposed explanation: “A **citation** proves every nearby **claim**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A generated report says 18 of 24 tests passed, so success was **80%**. Verify the **claim** in two ways and write a corrected sentence.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

An answer says six groups of four make 28.

Verify **independent**ly.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

An assistant says the school library opens at 8, but today's official notice says 9. Which **source** should guide today's visit? Explain the check.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 27 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A generated report says 18 of 24 tests passed, so success was **80%**. Verify the **claim** in two ways and write a corrected sentence.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss.

A draft says “M supplies 1250 g; the models use 900 g; 350 kg remains; every model will win a prize.” Check all four **claim**s, show the remaining mass in grams, and explain the status of the prize **claim**. Use 1 kg = 1000 g.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

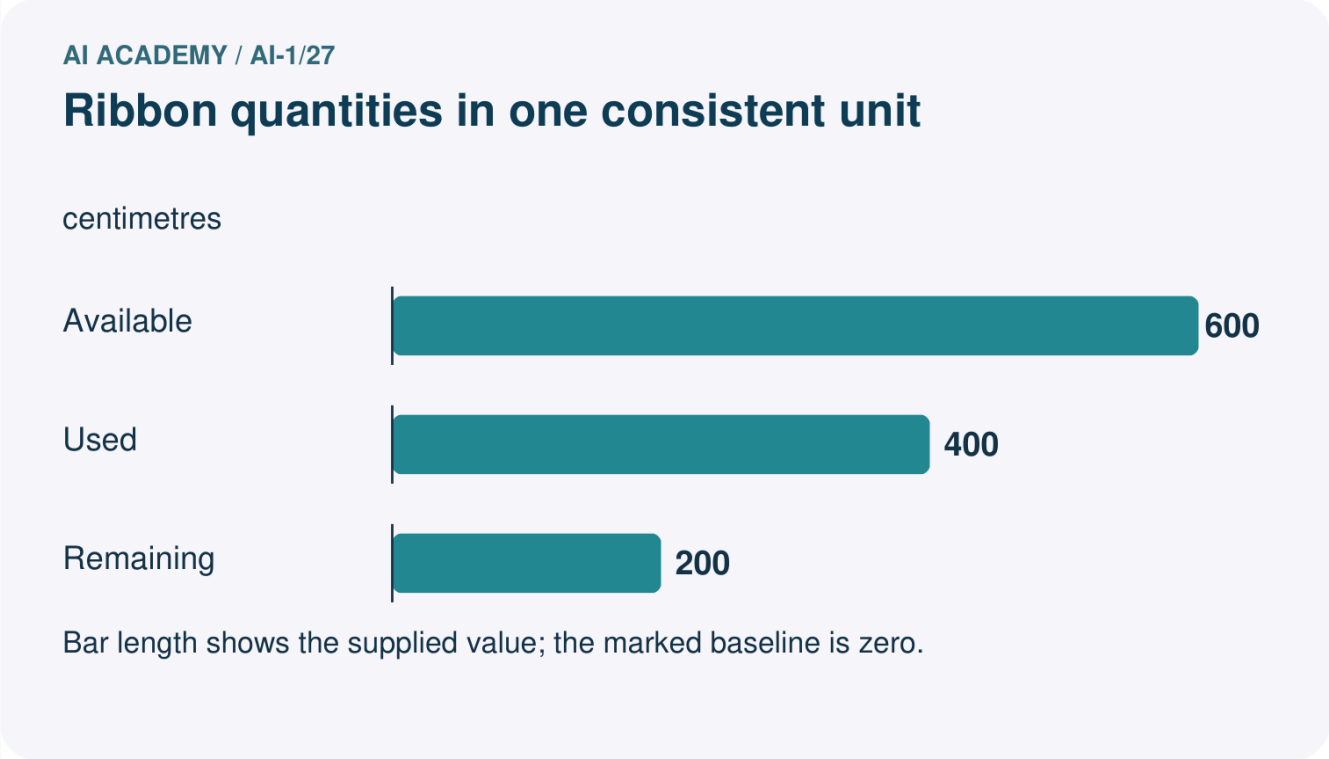
> Keep the explanation and the support you used.

Use this when helpful

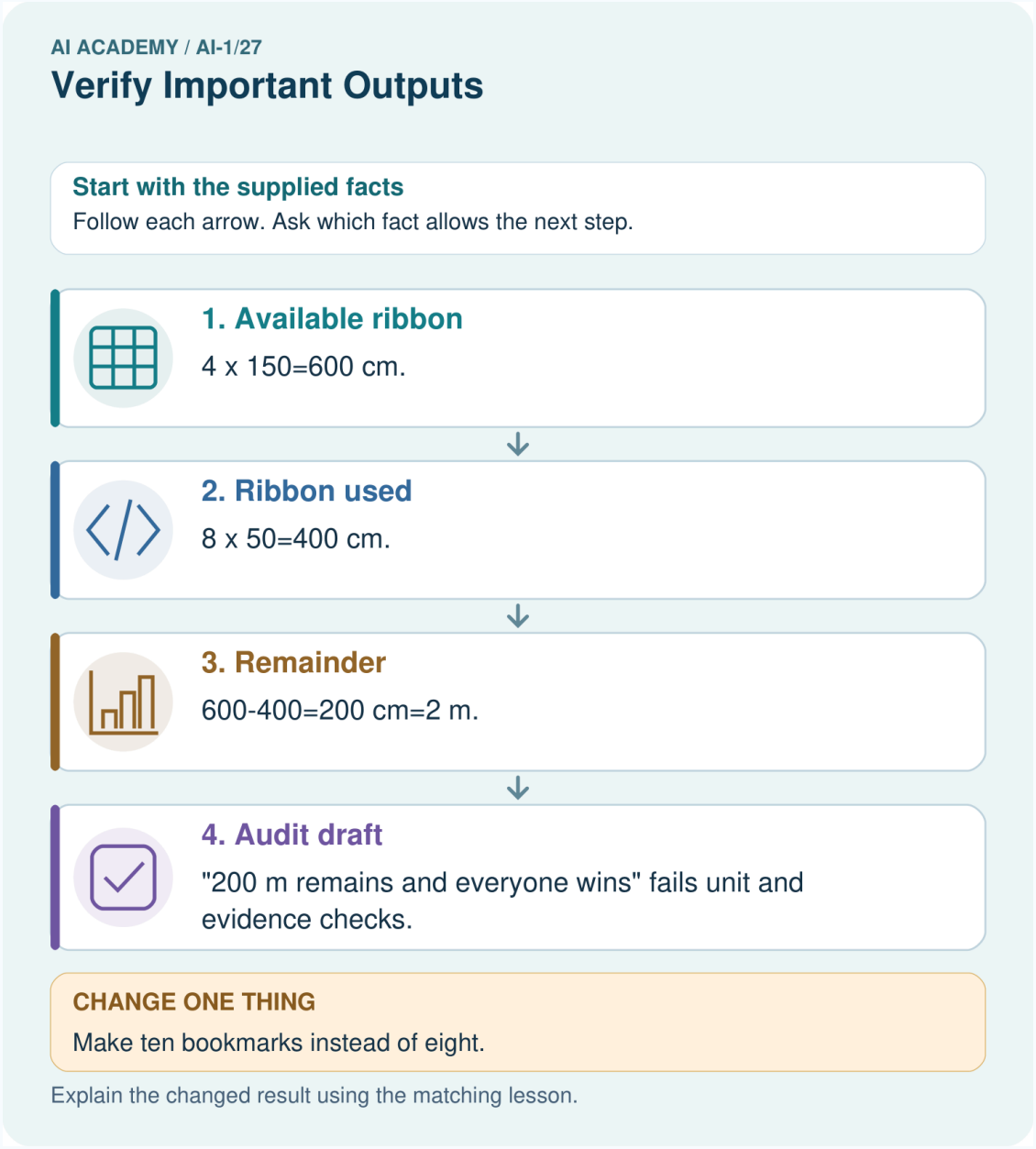
A fictional craft sheet calculates ribbon left after making bookmarks.

Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

No other ribbon loss is assumed; no competition result is recorded.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Correct the draft and identify which part needs another **source** rather than a calculation.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional craft sheet calculates ribbon left after making bookmarks.
- Which labelled step matches this detail? "200 m remains and everyone wins" fails **unit** and **evidence** checks.
- 1. Audit draft
- 2. Remainder
- 3. Ribbon used
- 4. Available ribbon
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Audit draft “200 m remains and everyone wins” fails **unit** and **evidence** checks. All arithmetic first uses centimetres.

Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**.

Changed-case question

- Changed-case practice: Make ten bookmarks instead of eight. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. 100 cm, or 1 m, remains.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: 100 cm, or 1 m, remains. 600−10×50=100 cm.

Repair and save your work

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, **unit** and meaning before using them. Short practice: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved **independent** assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

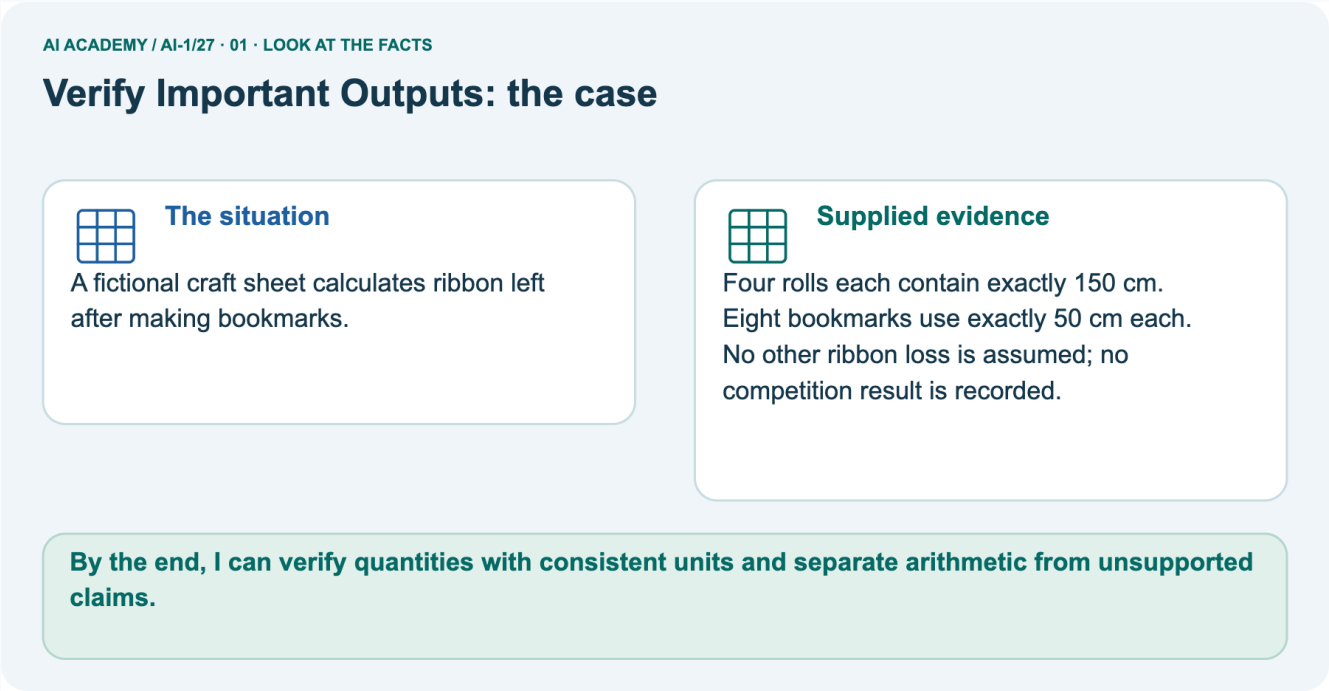
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Verify Important Outputs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

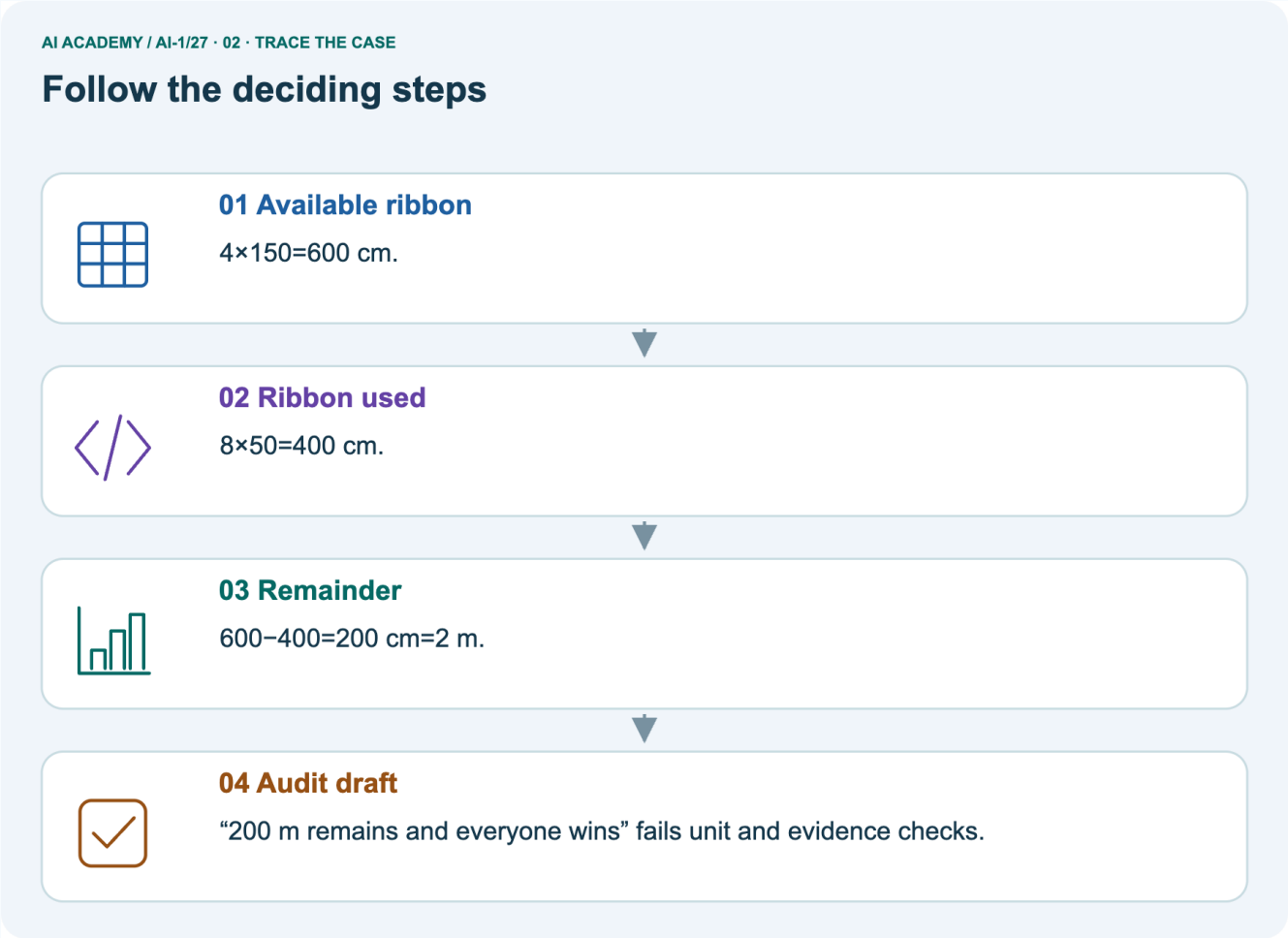
A fictional craft sheet calculates ribbon left after making bookmarks. Supplied facts: Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

No other ribbon loss is assumed; no competition result is recorded.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Available ribbon: 4×150=600 cm. → Ribbon used: 8×50=400 cm. → Remainder: 600−400=200 cm=2 m. → Audit draft: “200 m remains and everyone wins” fails **unit** and **evidence** checks.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Ribbon quantities in one consistent unit



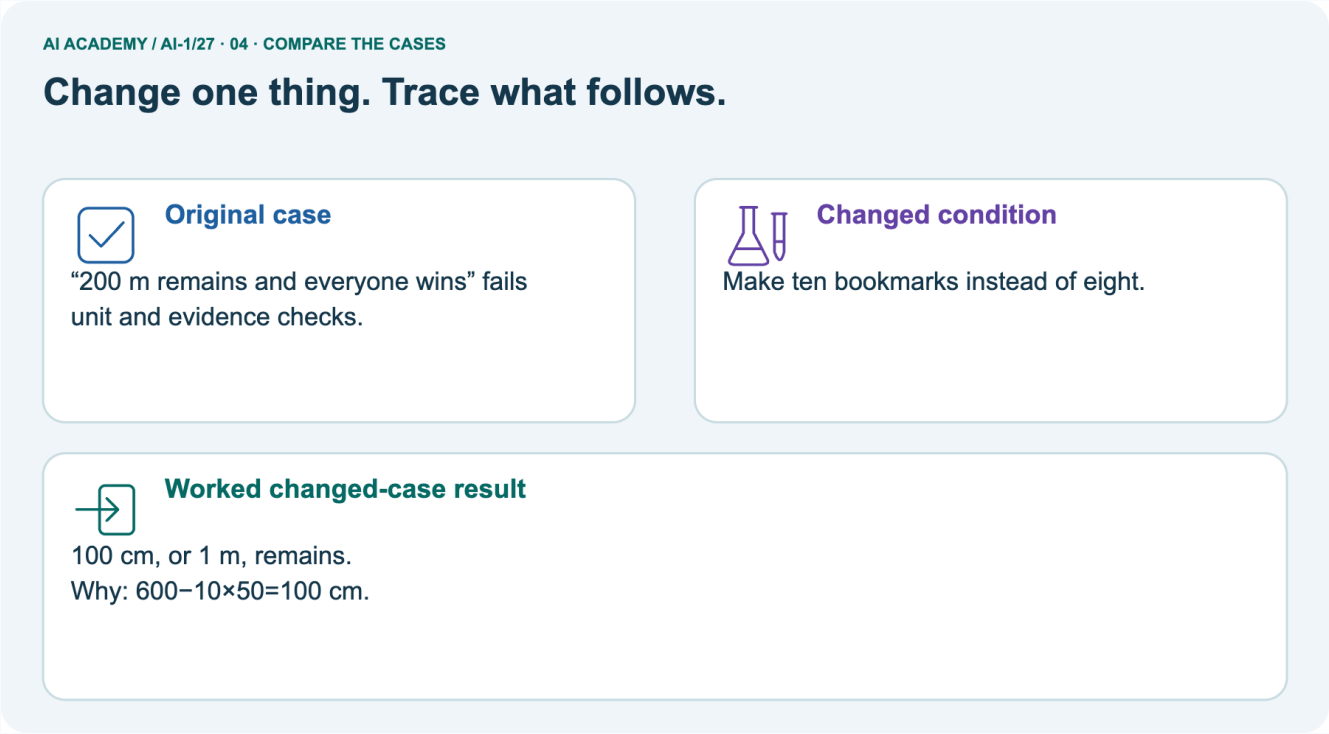
Follow the labelled picture with the matching explanation. Open the vector diagram

Ribbon quantities in one consistent **unit**. **Unit**s: centimetres. Available: 600. Used: 400. Remaining: 200.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



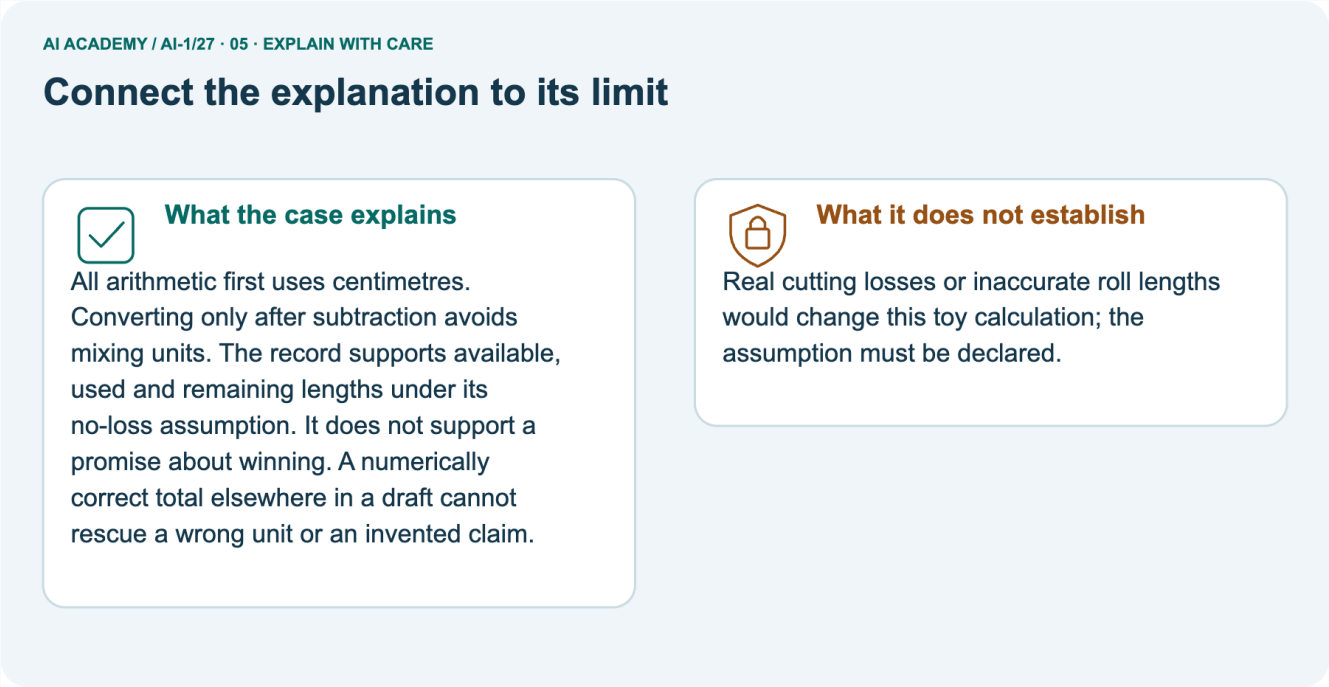
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: “200 m remains and everyone wins” fails **unit** and **evidence** checks. Changed condition: Make ten bookmarks instead of eight. Expected result: 100 cm, or 1 m, remains. Reason: 600−10×50=100 cm.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

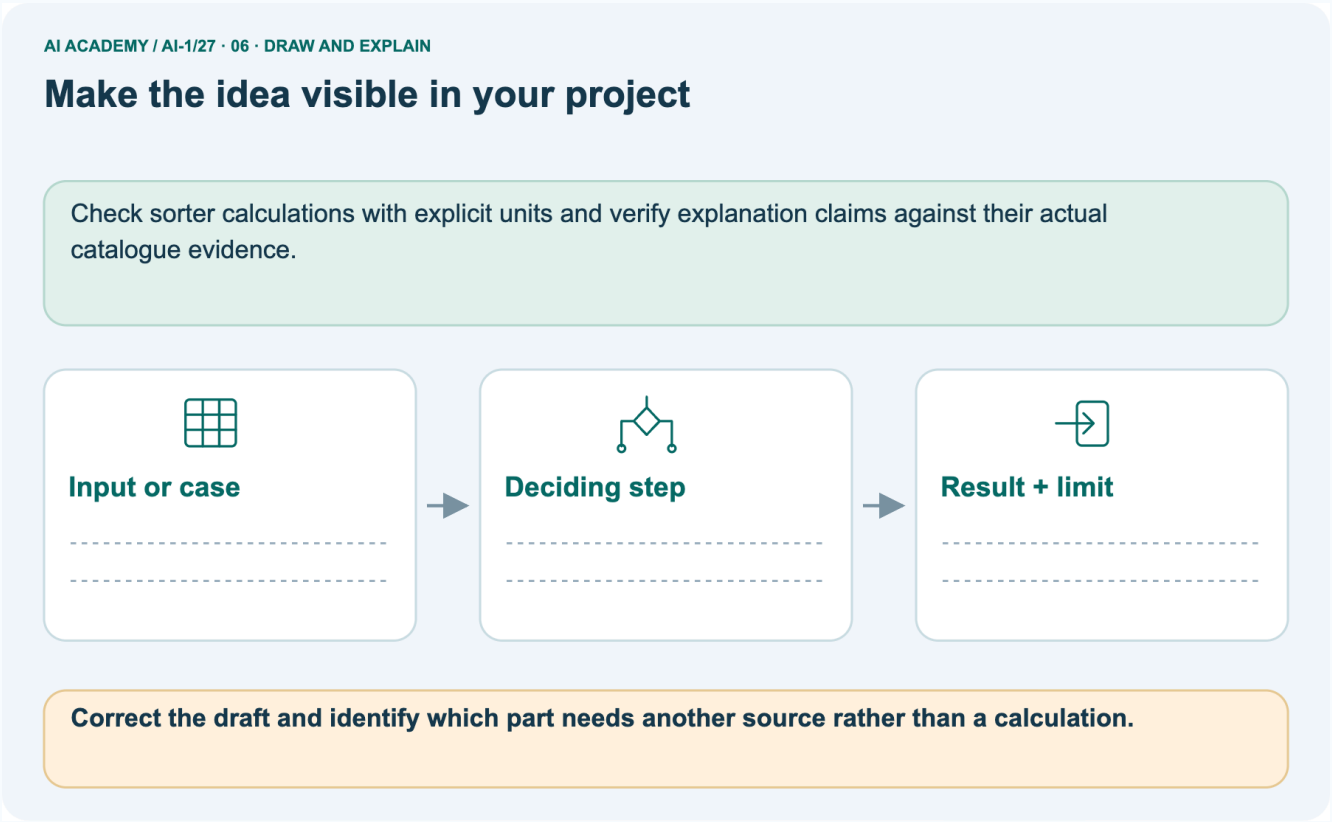
All arithmetic first uses centimetres. Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**. Limit: Real cutting losses or inaccurate roll lengths would change this toy calculation; the assumption must be declared.

Explain the case, then point to a limit on the **claim**.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Correct the draft and identify which part needs another **source** rather than a calculation.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Verify Important Outputs: the case

02. Follow the deciding steps

03. Ribbon quantities in one consistent unit

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.